

# MATLAB Assignment 3 [Output]

## Exercise 1

```
disp("==== Exercise 1 ====");
```

```
==== Exercise 1 ====
```

```
[name, ufid, ...  
  A, rref_A, det_A, det_AT, ...  
  A1, b1, sol_1_partic, sol_1_matlab, sol_1_cramer, ...  
  A2, b2, sol_2_partic, sol_2_matlab, sol_2_cramer, ...  
  A3, b3, sol_3_partic, sol_3_matlab, sol_3_cramer] = Exercise1()
```

```
Warning: Matrix is close to singular or badly scaled. Results may be inaccurate. RCOND = 6.167906e-18.
```

```
sol_3_partic = 3x1
```

```
4  
-2  
0
```

```
Warning: Matrix is close to singular or badly scaled. Results may be inaccurate. RCOND = 6.167906e-18.
```

```
sol_3_matlab = 3x1
```

```
1  
0  
-1
```

```
sol_3_cramer =
```

```
NaN
```

```
name =
```

```
"Evan Baesler"
```

```
ufid =
```

```
31151619
```

```
A = 3x3
```

```
1    2    3  
4    5    6  
-7   -8   -9
```

```
rref_A = 3x3
```

```
1    0   -1  
0    1    2  
0    0    0
```

```
det_A =
```

```
-6.6613e-16
```

```
det_AT =
```

```
0
```

```
A1 = 3x3
```

```
0    1    4  
1    3    3  
3    7    5
```

```
b1 = 3x1
```

```
-4  
-2  
6
```

```
sol_1_partic = 3x1
```

```
19  
-8  
1
```

```
sol_1_matlab = 3x1
```

```
19.0000  
-8.0000  
1.0000
```

```
sol_1_cramer = 3x1
```

```
19.0000  
-8.0000  
1.0000
```

```
A2 = 3x3
```

```

0      2      4
1      3      3
3      7      5
b2 = 3×1
-4
-2
6
sol_2_partic =

[]
sol_2_matlab = 3×1
1016 ×
1.8014
-1.2010
0.6005
sol_2_cramer =
NaN
A3 = 3×3
0      2      4
1      3      3
3      7      5
b3 = 3×1
-4
-2
-2
sol_3_partic = 3×1
4
-2
0
sol_3_matlab = 3×1
1
0
-1
sol_3_cramer =
NaN

```

```
disp("=====");
```

```
=====
```

## Exercise 2

```
disp("==== Exercise 2 ====");
```

```
==== Exercise 2 ====
```

```

[~, ~, ...
A1, b1, sol_1_partic, sol_1_matlab, sol_1_cramer, ...
A2, b2, sol_2_partic, sol_2_matlab, sol_2_cramer, ...
A3, b3, sol_3_partic, sol_3_matlab, sol_3_cramer] = Exercise2()

```

```

A1 = 6×4
2      5      0      -3
7      1      8      3
-1     8     -3      5
2     -3      3      2
-1     -2      7     -4
1      7      2     -4
sol_1_partic =

[]
sol_1_matlab = 4×1
-1.2781

```

```

0.3551
1.2209
0.4408
sol_1_cramer =
NaN
A2 = 4x6
    -3    -3     4    -1    -1    -4
     4     7     4     6    -1    -1
     7     2    -2    -2     4    -1
    -3    -4     7     4    -4     5
sol_2_partic = 6x1
    0.8165
   -0.2360
    1.6727
   -1.0508
     0
     0
sol_2_matlab = 6x1
    1.1075
   -1.0327
    1.0810
     0
     0
   -0.4751
sol_2_cramer =
NaN
sol_3_partic = 6x1
    0.5471
    0.5745
   -0.0706
   -1.0719
   -0.9839
   -0.2278
sol_3_matlab = 6x1
    0.5471
    0.5745
   -0.0706
   -1.0719
   -0.9839
   -0.2278
sol_3_cramer = 6x1
    0.5471
    0.5745
   -0.0706
   -1.0719
   -0.9839
   -0.2278
A1 = 6x4
     2     5     0    -3
     7     1     8     3
    -1     8    -3     5
     2    -3     3     2
    -1    -2     7    -4
     1     7     2    -4
b1 = 6x1
    -4
     4
     2
    -3
     7
     2
sol_1_partic =

[]
sol_1_matlab = 4x1
   -1.2781

```

```

0.3551
1.2209
0.4408
sol_1_cramer =
NaN
A2 = 4x6
    -3    -3     4    -1    -1    -4
     4     7     4     6    -1    -1
     7     2    -2    -2     4    -1
    -3    -4     7     4    -4     5
b2 = 4x1
     6
     2
     4
     6
sol_2_partic = 6x1
    0.8165
   -0.2360
    1.6727
   -1.0508
     0
     0
sol_2_matlab = 6x1
    1.1075
   -1.0327
    1.0810
     0
     0
   -0.4751
sol_2_cramer =
NaN
A3 = 6x6
    -2     1     8    -3     6     1
     1     5     6    -3     1     1
    -3     2     5     0    -1     5
     6     1     4    -3     2     8
     6     1     8     1     6    -3
     3    -3     1     5    -2     2
b3 = 6x1
    -4
     5
    -1
     3
    -3
    -4
sol_3_partic = 6x1
    0.5471
    0.5745
   -0.0706
   -1.0719
   -0.9839
   -0.2278
sol_3_matlab = 6x1
    0.5471
    0.5745
   -0.0706
   -1.0719
   -0.9839
   -0.2278
sol_3_cramer = 6x1
    0.5471
    0.5745
   -0.0706
   -1.0719
   -0.9839
   -0.2278

```

```
disp("=====");
```

```
=====
```

## Exercise 3

```
disp("==== Exercise 3 ====");
```

```
==== Exercise 3 ====
```

```
[~, ~, ...  
 N1, B1, pivcols1, C1, R1, ...  
 N2, B2, pivcols2, C2, R2, ...  
 N3, B3, pivcols3, C3, R3, ...  
 A4, N4, B4, pivcols4, C4, R4, ...  
 rank_A1, rank_A2, rank_A3, rank_A4] = Exercise3(A1, A2, A3)
```

```
N4 = 6×3
```

```
0.9359    0.2469   -0.1753  
0.0339    0.3946   -0.1858  
-0.3370    0.5202   -0.6975  
-0.0925    0.6857    0.6412  
0.0231   -0.1714   -0.1603  
-0.0154    0.1143    0.1069
```

```
C4 = 4×3
```

```
1     0     -4  
0     1     4  
-1    0     0  
2    -1   -10
```

```
N1 =
```

```
4×0 empty double matrix
```

```
B1 = 6×4
```

```
1     0     0     0  
0     1     0     0  
0     0     1     0  
0     0     0     1  
0     0     0     0  
0     0     0     0
```

```
pivcols1 = 1×4
```

```
1     2     3     4
```

```
C1 = 6×4
```

```
2     5     0    -3  
7     1     8     3  
-1     8    -3     5  
2    -3     3     2  
-1    -2     7    -4  
1     7     2    -4
```

```
R1 = 4×4
```

```
1     0     0     0  
0     1     0     0  
0     0     1     0  
0     0     0     1
```

```
N2 = 6×2
```

```
-0.3545   -0.2332  
-0.0138    0.5166  
0.0888    0.3959  
0.3473   -0.6406  
0.8612    0.1066  
0.0630    0.3168
```

```
B2 = 4×6
```

```
1.0000     0     0     0    0.3669    0.6124  
0     1.0000     0     0    0.1387   -1.6770
```

```

      0      0      1.0000      0      -0.0120      -1.2455
      0      0      0      1.0000      -0.5650      2.2119
pivcols2 = 1x4
      1      2      3      4
C2 = 4x4
      -3      -3      4      -1
      4      7      4      6
      7      2      -2      -2
      -3      -4      7      4
R2 = 4x6
      1.0000      0      0      0      0.3669      0.6124
      0      1.0000      0      0      0.1387      -1.6770
      0      0      1.0000      0      -0.0120      -1.2455
      0      0      0      1.0000      -0.5650      2.2119
N3 =

      6x0 empty double matrix
B3 = 6x6
      1      0      0      0      0      0
      0      1      0      0      0      0
      0      0      1      0      0      0
      0      0      0      1      0      0
      0      0      0      0      1      0
      0      0      0      0      0      1
pivcols3 = 1x6
      1      2      3      4      5      6
C3 = 6x6
      -2      1      8      -3      6      1
      1      5      6      -3      1      1
      -3      2      5      0      -1      5
      6      1      4      -3      2      8
      6      1      8      1      6      -3
      3      -3      1      5      -2      2
R3 = 6x6
      1      0      0      0      0      0
      0      1      0      0      0      0
      0      0      1      0      0      0
      0      0      0      1      0      0
      0      0      0      0      1      0
      0      0      0      0      0      1
A4 = 4x6
      1      -5      2      0      -4      0
      0      0      0      1      4      0
      -1      5      -2      0      0      -6
      2      -10      4      -1      -10      3
N4 = 6x3
      0.9359      0.2469      -0.1753
      0.0339      0.3946      -0.1858
      -0.3370      0.5202      -0.6975
      -0.0925      0.6857      0.6412
      0.0231      -0.1714      -0.1603
      -0.0154      0.1143      0.1069
B4 = 4x6
      1.0000      -5.0000      2.0000      0      0      6.0000
      0      0      0      1.0000      0      -6.0000
      0      0      0      0      1.0000      1.5000
      0      0      0      0      0      0
pivcols4 = 1x3
      1      4      5
C4 = 4x3
      1      0      -4
      0      1      4
      -1      0      0
      2      -1      -10
R4 = 3x6
      1.0000      -5.0000      2.0000      0      0      6.0000

```

```
      0      0      0      1.0000      0      -6.0000
      0      0      0      0      1.0000      1.5000
rank_A1 =
4
rank_A2 =
4
rank_A3 =
6
rank_A4 =
3
```

```
disp("=====");
```

```
=====
```